

was sent in the packet. If these don't match, the computer figures that the data has been corrupted en route, discards the packet, and requests the sender to re-send the packet. TCP then assembles them into their original coherent form to allow the receiving computer to make sense of it.

None of this would be possible, however, without a DNS server acting as mediator.

2.1.10 The Domain Name System

Most of us would find it considerably difficult to remember hundreds of strings of numbers of all the sites we visit. So, for humans, the address is in letters (the URL). The domain name is the root identifier of a Web site on the Internet.

When you want to contact a location on the Internet—say a Web site—you'd type in an address like **www.yahoo.com**. IP uses (a) the Internet address information and (b) the Domain Name System to understand and deliver your mail or request from one computer to another: the letters **www.yahoo.com** are translated into an IP address, which is, in this case, 87.248.113.14. In fact, if you typed in `http://87.248.113.14`, your screen would show you the Yahoo! site, and you would make the job a microsecond faster and a trifle easier for the computers. This translation is accomplished by the DNS server.

Before we understand what the DNS server does, let's get a grip of how the DNS works.

2.1.11 How The Domain Name System Works

In order to make efficient and hassle-free the translation from the plain-English **www.yahoo.com** into numbers that computers can understand, the Internet has been organised into a number of major domains. Major domains are represented by the letters at the end of the English address. These are called top-level domains (TLD) or zones, and they are either two or three letters long. The three-letter zones indicate the type of organisation that owns the domain, that is, whether it's a commercial institution, ISP, non-governmental organisation of academic establishment, etc. All Internet domains are registered with private companies called Internet Registrars, in co-ordination with the InterNIC (Internet